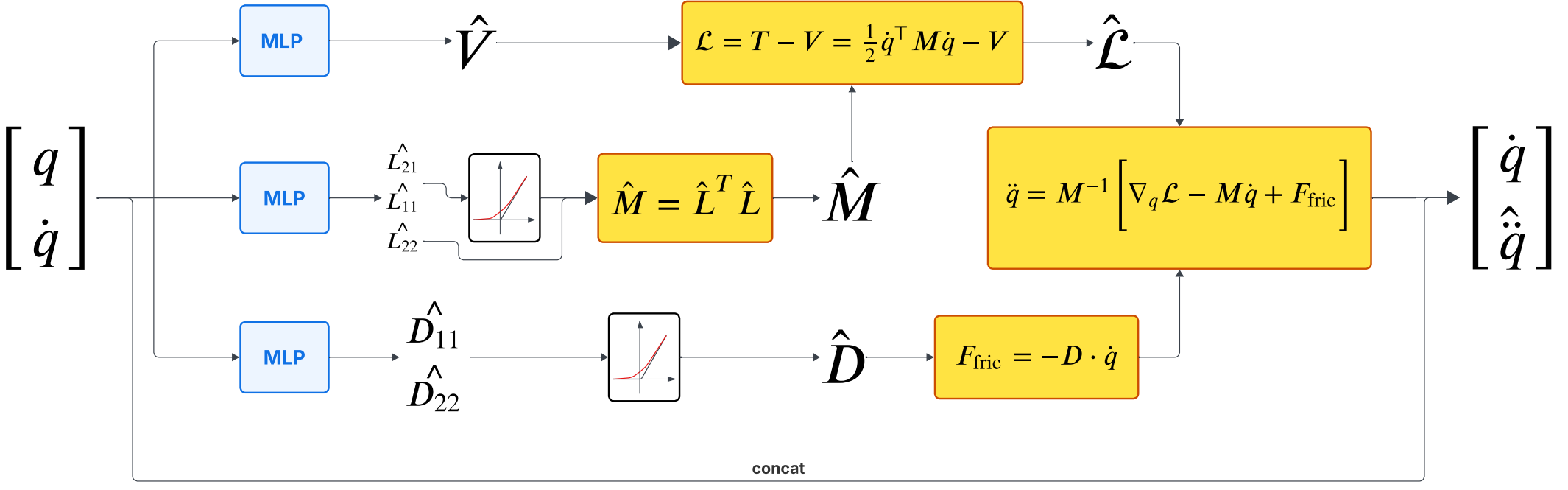
$\begin{bmatrix} q \\ \dot{q} \end{bmatrix}$
MLP
MLP
MLP
$\hat{V}$
$\mathcal{L} = T - V = \frac{1}{2}\dot{q}^\top M \dot{q} - V$
$\hat{\mathcal{L}}$
$\hat{L}_{21}$
$\hat{L}_{11}$
$\hat{L}_{22}$
$\hat{M} = \hat{L}^T \hat{L}$
$\hat{M}$
$\hat{D}_{11}$
$\hat{D}_{22}$
$\hat{D}$
$F_{\text{fric}} = -D \cdot \dot{q}$
$\ddot{q} = M^{-1}\left[\nabla_q \mathcal{L} - M\dot{q} + F_{\text{fric}}\right]$
$\begin{bmatrix} \dot{q} \\ \ddot{q} \end{bmatrix}$
concat